

Ensembles and Random Forests

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Limitations of Decision Trees

- ▶ Decision trees are easy to interpret and don't require much computation to train
 - ▶ However, capturing a complex relationship using a decision tree requires the tree be deep (lots of splits)
 - ▶ This is problematic because trees exhibit high variance and are prone to overfitting
- ▶ This presentation will focus on the *random forest* algorithm, a well-known *ensemble model*

Bagging

Random forests rely upon *bagging*, or “bootstrap aggregation”, to construct a large number of decision trees:

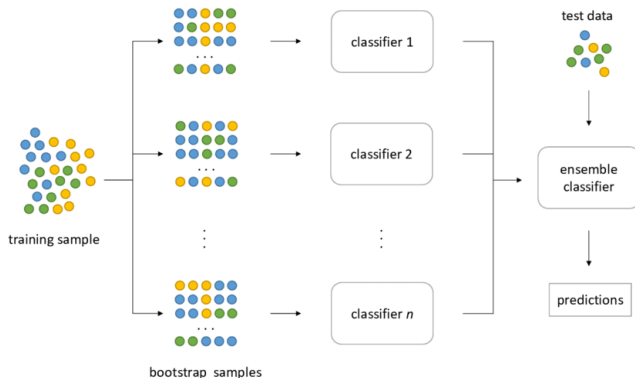


Image credit: <https://hudsonthames.org/bagging-in-financial-machine-learning-sequential-bootstrapping-python/>

- ▶ Bagging produces an *ensemble model* comprised of many different *base models*
 - ▶ Each base model contributes towards a final prediction, either by majority/weighted voting (classification) or simple/weighted averaging (regression)

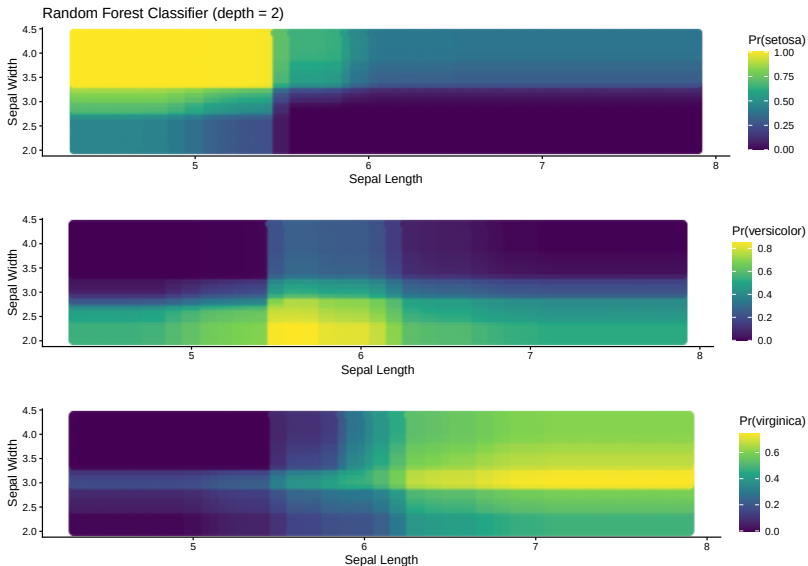
Ensembles

- ▶ Bagging produces an *ensemble model* comprised of many different *base models*
 - ▶ Each base model contributes towards a final prediction, either by majority/weighted voting (classification) or simple/weighted averaging (regression)
- ▶ Random forests are an ensemble model built using bagging, where each base model is a decision tree
 - ▶ What would happen if bagging were not used?

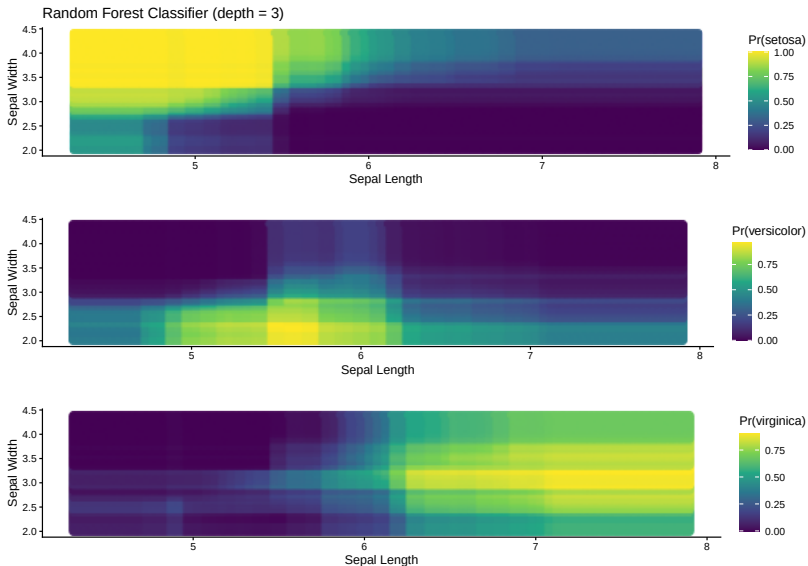
Random Forest

- ▶ Bagging is one strategy used by random forests to address the limitations of a single decision tree
- ▶ A second strategy is *predictor sampling*, or the random selection of limited candidate pool of predictors to be considered *at each split*
 - ▶ What might happen if predictor sampling were not used?

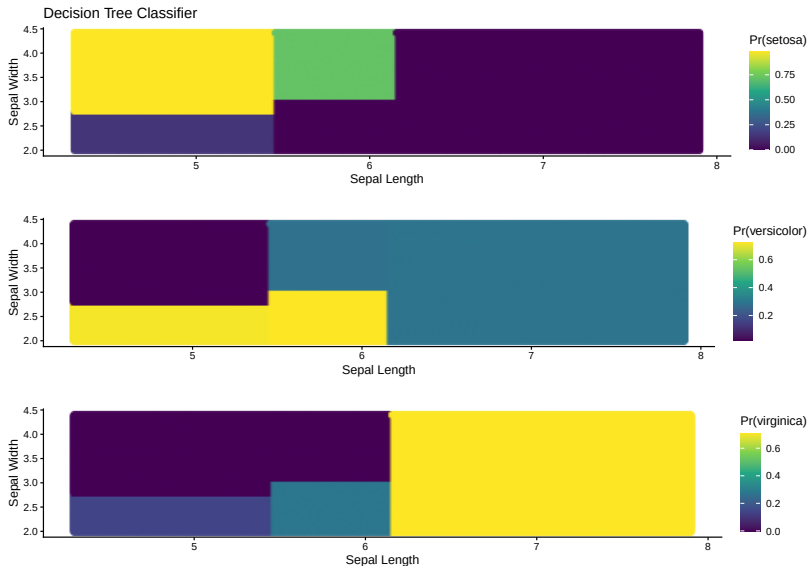
Random Forest (depth = 2)



Random Forest (depth = 3)



Single Decision Tree (depth = 3)



Final Remarks

- ▶ Random forests will generally offer better predictive performance than a single decision tree
 - ▶ The primary downside is that random forests are not easily interpretable
- ▶ Important tuning parameters are `max_depth`, `min_samples_split`, and `max_features` (the number or fraction of predictors considered at each split)
- ▶ The number of trees in the forest is also important, but including more trees past a certain point will not improve the ensemble.

Final Remarks (cont.)

- ▶ `max_depth` and `min_samples_split` help prevent base models from being overfit
 - ▶ By using an ensemble approach, random forests can be flexible without using deep trees
 - ▶ Thus, relative to a single decision tree, you should consider using a smaller `max_depth` and larger `min_samples_split`
- ▶ `max_features` governs the degree of correlation between base models
 - ▶ Smaller values reduce correlations between trees (at the expense of predictive power within individual trees)
 - ▶ Default recommendations are $\sqrt{p}$ for classification and $p/3$ for regression